

IAN SNIDER

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Robotist designing autonomous aerial vehicles.

EDUCATION

Ph.D. Nuclear Engineering - Robotics - University of California, Berkeley		Aug 2025 - present
B.S. Mechanical Engineering - Washington University in St. Louis	4.00/4.00	May 2025
B.A. Physics, Math minor - Truman State University	3.89/4.00	Dec 2024

PROJECTS

TARES Tethered Aerial Radiation Estimation System - Researcher Jan 2026 - present

- Designed a UAV system for controlling the trajectory of a tether-suspended payload
- Flew autonomous tests at Richmond Field Station and Lawrence-Berkeley under Part 107 regulations

Micro Mocap Low-Cost Mocap System for Small Drones - Developer Summer 2026

- A motion capture state estimation system using PS3eye cameras for tracking IR leds
- ~\$10 per camera and an ~\$80 custom racing drone system that achieved stable maneuvering

Sniff Consensus Protocol for Noisy Multi-Agent Formations - Developer Fall 2025

- Consensus protocol for a random graph of agents perturbed with communication noise

WashU Robotics MateROV & MARINER Projects - Mechanical Lead 2023 - 2025

- Designed and tested an autonomous underwater vehicle (AUV)
- Prototyped dive control using an IMU and depth state estimation with a Kalman Filter for 1D hovering
- Built a chassis and buoyancy engine

CrazyFlie 2.1 cascaded PID/PD Control System - ESE 4481 Student Spring 2025

- Designed a control system for a CrazyFlie 2.1 quadcopter and achieved stable autonomous flight

EXPERIENCE

Lawrence Berkeley National Laboratory - Lab Affiliate, Berkeley, CA Aug 2025 - present

- Developed a method for a hexacopter to control the trajectory of a 10m tether-suspended radiation detector
- Designed and implemented safety-critical software infrastructure for autonomous systems and outer-loop control algorithms for commanding ArduPilot inner-loop controller
- Architected a communication pipeline for GCS, transmitter, companion computer, and Cube Blue flight controller
- Executed autonomous flight tests at LBNL and Richmond Field Stations under Part 107

Lawrence Livermore National Laboratory - Graduate Intern, Livermore, CA June 2025 - Dec 2025

- Developed CRISP (CRITICAL Simulation Pipeline), a tool for distributing high-performance computing (HPC) jobs
- CRISP successfully automated compilation and benchmarking workflows for ~3420 reactor simulation cases

Brookhaven National Laboratory - Student Researcher, Upton, NY Summers 2022 - 2024

- Developed BRR (Bayesian Resonance Reclassifier) for reclassification tasks on HPC clusters

Truman State University - Student Astronomy Researcher, Kirksville, MO 2021 - 2022

- Calculated trajectories for ~3000 Starlink satellites to optimize telescope viewing plans and developed a GUI

SKILLS

- **Programming:** C, C++, Python, Go, Bash, Lua, MATLAB, Mathematica, Octave, LaTeX, Typst, Vimscript
- **Software/Technical:** MAVLink, ROS2, Simulink, SolidWorks, Scikit-learn, Slurm, Git, Linux
- **Physics/engineering:** Autonomous Aerial Vehicle Control, Classical Mechanics, Electrodynamics, Vibrations, Mathematical Physics, Fluid Mechanics, Solid Mechanics, Acoustics, Materials Science, Thermal Systems, Propulsion
- **Math:** Linear Algebra, ODEs, Computing Structures, Control Systems, Machine Learning, Random Walks, Random Matrix Theory, and Optimizations